FROM FAILURE TO SUCCESS

How the lessons learnt from the 2019 Chandrayaan-2 disaster contributed to the success of India's third lunar exploration mission

Graphic by **NILANJAN DAS** *and* **TANMOY CHAKRABORTY**

CHANDRAYAAN-3: MISSION OBJECTIVES

- To demonstrate safe and soft landing on lunar surface–*Done*
- To demonstrate rover traversing the moon–*Done*
- To conduct in-situ scientific experiments –*In progress*

LANDING RELIABILITY

- Integrated hot test for close-up system assessment of engines, sensors, navigation, guidance and control
- Integrated cold test to thoroughly assess all sensors and navigation systems
- Lander-led drop tests to assist the touchdown capabilities
- Landing leg mechanism strengthened for higher impact
- Landing site expanded by five times, from 500m x 500m to 4km x 2.4km
- Complete redundancies of all systems built into lander





